

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

MAY EXAMINATION

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS

MONDAY, 22 MAY 2023

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGY

Instruction to Candidates:

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL THREE (3)** questions in **Section A** and choose **ONLY ONE (1)** question in **Section B**.

Should a candidate answer more than **ONE (1)** question in section B, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are not allowed to use calculator.

Answer questions only in the answer booklet provided.

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Section A [Total: 75 marks]**

This section consists of **THREE (3) COMPULSORY** questions.

- Q1. (a) The relational schema shown below is part of a hospital management database. The primary keys are highlighted in bold.

Patient (**patientID**, patName, patAddress, patDOB, guardianName, guardianPhone)

Ward (**wardNo**, wardType, noOfBeds)

Admission (**patientID**, **wardNo**, admitStartDate, admitEndDate)

Medicine (**medNo**, medName, costPerUnit)

Prescription (**patientID**, **medNo**, unitPerDay, startDate, finishDate)

- (i) Write **ONE (1)** Structured Query Language (SQL) statement to list all the ward type in alphabetical order. (3 marks)
 - (ii) Write **ONE (1)** SQL statement to identify all patients name admitted to "Orthopedic" ward in 2022. (6 marks)
 - (iii) Write **ONE (1)** SQL statement to calculate the total cost of "Aspirin" supplied to patient name "Steven Liu". Name the output as "Medical Cost". (7 marks)
- (b) The tables below contain some data for a hypermarket that has multiple branches in different states throughout Malaysia.

Table 1.1 Partial records of Branch relation

branchCode	branchName	branchSales	stateCode
B05	Kenanga Hypermarket	RM1.3m	PK
B07	Azalea Hypermarket	RM1.9m	PK
B08	Laurel Hypermarket	RM1.1m	KD
B06	Aster Hypermarket	RM2.2m	TG
B02	Gerbera Hypermarket	RM1.3m	KD

Table 1.2 Partial records of Employee relation

empCode	empName	empDOB	branchCode	jobCode
E0023	Ahmad	25-Feb-1993	B08	JC33
E0100	Thomas	18-Apr-1987	B07	JC22
E0078	Julia	30-Jan-1992	B05	JC11
E0003	Katie	19-May-1985	B06	JC22
E0202	Kamal	14-Sep-1992	B08	JC11
E0099	Suraya	23-Aug-1985	B06	JC11

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q1. (b) (Continued)**

Table 1.3 Partial records of Job relation

jobCode	jobDesc	jobSalary
JC11	Cashier	RM1700
JC22	Manager	RM2500
JC33	Butcher	RM1500

Table 1.4 Partial records of State relation

stateCode	stateName
PK	Perak
KD	Kedah
TG	Terengganu

- (i) Write **ONE (1)** SQL statement to calculate the total sales of all branches in every state of Malaysia. Display the total sales with the code of each state. (3 marks)

Table 1.5 The expected output of total sales by state

stateCode	Total Sales
PK	RM3.2m
KD	RM2.4m
TG	RM2.2m

- (ii) Write **ONE (1)** SQL statement to shows 25 % of increment to the salary of all employees who work in the state of Kedah. Display the job code, job description, original salary and the new salary. (6 marks)
[Total : 25 marks]

- Q2. (a) The following is an entity relationship diagram (ERD) of the product purchased by some customers in a stationary shop. Each sales order is managed by a particular employee of the stationary shop.

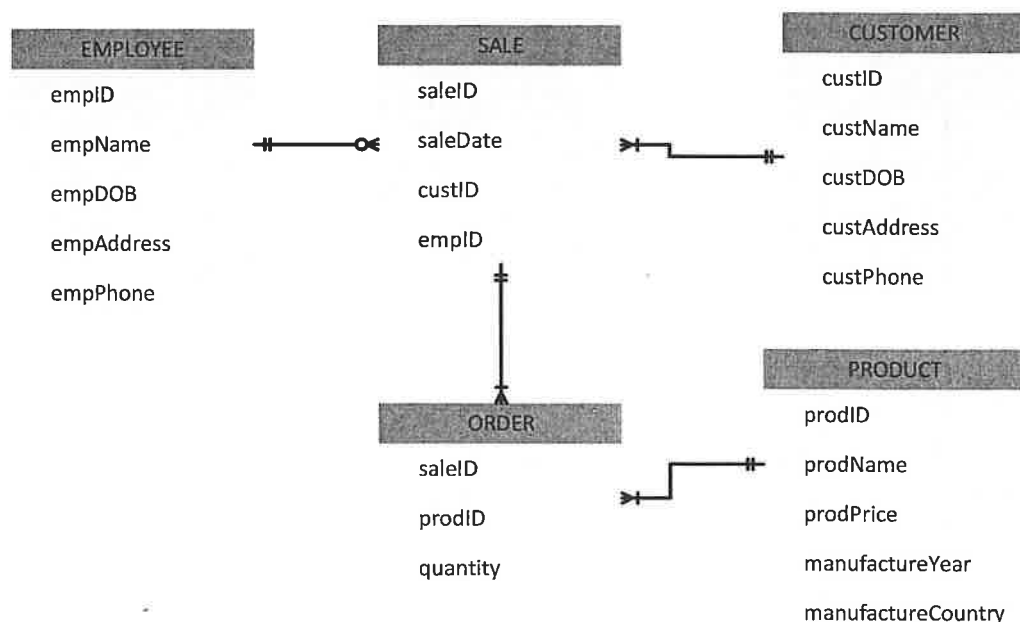


Figure 2.1 Product Purchase Order

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q2. (a) (Continued)**

- (i) Suggest an appropriate data type (Oracle standard) for the *empAddress* and *quantity* attributes from Employee and Order entities respectively. (2 marks)
- (ii) Identify **ONE (1)** existence-dependent and **ONE (1)** existence-independent entity from the above ERD. Justify your answer. (6 marks)
- (iii) Identify and explain **ONE (1)** associate entity in Figure 2.1. (3 marks)
- (iv) Explain the integrity constraint(s) applied to Sale entity. (6 marks)
- (b) Explain the terms 'Data Redundancy' and 'Data Inconsistency'. (4 marks)
- (c) State **FOUR (4)** actions that can be applied if there exists a child occurrence referencing a parent occurrence that we wish to delete. (4 marks)
- [Total : 25 marks]

- Q3. (a) The hotel manager wants to have an effective information system that can help him plan meals and keep track of who attends lunch sessions at the hotel. Currently, information is stored in a table as in Table 3.1. Customers may attend multiple lunch sessions but will not attend more than one lunch session on the same date. They can also order various foods that they want during the session.

Table 3.1 Sample Records for the Lunch Session

CUSTOMER NUM	CUSTOMER NAME	CUSTOMER ADDRESS	LUNCH NUM	LUNCH DATE	VENUE CODE	VENUE DESCRIPTION	FOOD CODE	FOOD DETAILS
C001	James	303 Cambrigde Road	L0045	14-Mar-2023	011	Deli Cafe	MD1	Grilled Lamb
							SA3	Spinach Salad
C088	Edward	577 Orchard Road	L0100	23-Feb-2023	012	Grand Ballroom	MD2	Smoked Chicken
							S03	Spinach Salad
C071	Rose Mary	101 Campbell St.	L0009	23-Feb-2023	050	Lexus Sulte	MD1	Grilled Lamb
							AP7	Shrimp Tacos
							AP4	Mushroom Soup
C100	Louis	321 Victoria Ct.	L0100	23-Feb-2023	012	Grand Ballroom	MD5	Carnonara Pasta
C088	Edward	577 Orchard Road	L0188	19-Mar-2023	050	Lexus Suite	MD1	Grilled Lamb
							AP4	Mushroom Soup

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q3. (a) (Continued)**

- (i) Convert the table form Unnormalized Form to First Normal Form (1NF). (3 marks)
 - (ii) Based on the answer in Q3(a)(i), identify the primary key of the relation. (2 marks)
 - (iii) Based on the answer in Q3(a)(ii), draw the dependency diagram showing all the dependencies in the relation. (6 marks)
 - (iv) Decompose the table in Table 3.1 into relations that are in Third Normal Form (3NF). The resulting relations should retain all the information in the original table. (10 marks)
- (b) Figure 3.2 shows the sample of dependency diagram. Identify all the determinant(s). (4 marks)

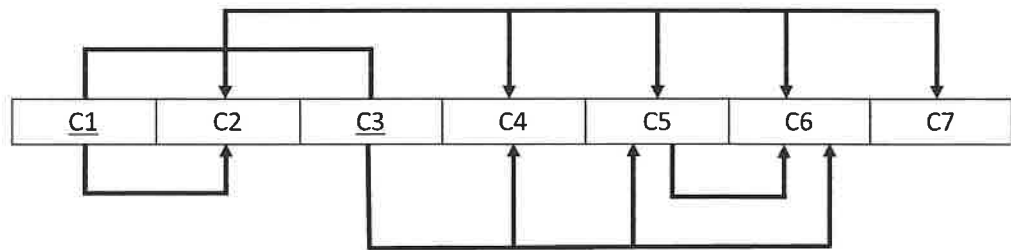


Figure 3.2 Dependency Diagram

[Total : 25 marks]

Section B [Total: 25 marks]

This section consists of **TWO (2) OPTIONAL** questions. Answer **ONE (1)** question only.

- Q4. (a) Figure 4.1 shows the partial of entity relationship diagram (ERD) for company vehicle management database.

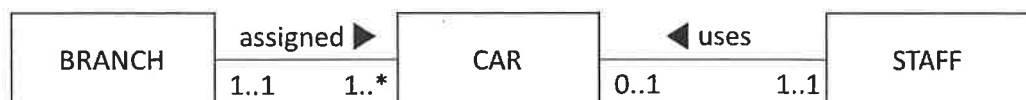


Figure 4.1 Partial ERD for Company Vehicle Management Database

- (i) Identify and describe the connection trap that occurred. (3 marks)
- (ii) Give an example of situation where the problem may arise from querying the ERD as in Figure 4.1. (3 marks)

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q4. (Continued)**

- (b) The relational schema shown below is part of Danish Academy database. Note that the primary key is underlined.

Student (stuID, Name, Address, DOB, Gender, PhoneNo)

Course (courseCode, courseName, creditHr)

Enrollment (stuID, courseCode, Grade)

CourseOffering (OfferNo, OfferSemester, OfferYear, OfferDay, OfferTime)

Enrollment.stuID is the foreign key that references Student.stuID

Enrollment.courseCode is the foreign key that references Course.courseCode

There is a flaw in the design causing some of the required information to not be stored. Here are some of the weaknesses detected.

- The database can currently only capture the final grade for any course taken by any student.
- The design cannot store information about when a student registers for a course.

Suggest **ONE (1)** way to fix the design flaw by illustrating it into a complete ERD. (4 marks)

- (c) Atlantis Real Estate Company (ARE) is a real estate company and has several sales offices in several states. In order to increase the effectiveness of data management, the company has hired your company to build a database. As a database developer, you are tasked with designing the database based on the data requirements that have been collected. The sales office information that needs to be recorded is the sales office id, address, city, state, zip code, phone number and fax number. For each sales office, at least 20 employees will be assigned. Each employee will be assigned to one sales office only and will be supervised by a manager. The information about the employee that needs to be recorded is the employee number, name, position, address, start date, supervisor number, phone number and mobile phone number. Regarding the company's operations, each sales office has a list of properties for sale. Each property available for sale will be managed by one sales office only. Each property needs to store information such as address, city, state, zip code, property type, property code and property sale price. Each property can have several nominees as its owner. Each owner will be given an id number and all the properties he owns will be listed. In addition, each property will have information about the percentage of ownership by its owner.

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q4. (c) (Continued)**

Draw an ERD from the above scenario that includes all the entities, attributes of each entity, relationships, primary key, foreign key(s) and the multiplicity of the relationship, i.e., cardinality and optionality. For each entity, list at least three relevant attributes except for those which have been assigned. Use **CROW'S FOOT notation** or **UML notation** to illustrate your ERD.

(15 marks)

[Total : 25 marks]

- Q5. (a) To increase efficiency of daily operations, ABC Enterprise plans to automate the process of managing staff information. After a series of requirement gathering activities, as a System Analysis, you are required to come up with an entity relationship diagram (ERD) showing the overall relationships of data. An employee may be assigned to different projects from time to time. For each project, there will be a manager and it needs at least two employees as members and should operate efficiently with about ten members. To total assign-hour per member is recorded upon completion of each project. It is important that a member is not assigned to the same project again if he/she leaves or is being terminated half-way through the project. While working in the company, an employee will receive at any one-time, different types of earnings such as basic salary, housing allowance, manager allowance etc. Some earnings are fixed for all employees while some are tailored to each individual. As such, the amount earned for each earning type of each employee is recorded. There is possibility that an employee may receive the same allowance again in future. For example, Encik Kamal holds the position of the manager for IT department from Jan 2020 until Dec 2020 and is entitled to RM1000 as manager allowance. In Jan 2023, he may be assigned the same post again due to his excellent track records and enjoys the same allowance amount.

Draw an ERD from the above scenario that includes all the entities, attributes of each entity, relationships, primary key, foreign key(s) and the multiplicity of the relationship, i.e., cardinality and optionality. For each entity, list at least three relevant attributes except for those which have been assigned. Use **CROW'S FOOT notation** or **UML notation** to illustrate your ERD.

(15 marks)

- (b) In factory A, there are three duty shifts which are morning, afternoon and night. Each employee is assigned to work in several duty shifts that rotate each day. Each employee is identified by an employee id and has a name, phone number and age. For each duty shift, a maximum of 150 workers will be allocated.
- (i) Identify the relationship type and the degree of the relationship.

(2 marks)

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q5. (b) (Continued)**

- (ii) Illustrate the above scenario with a complete ERD. Fix any issues that exist in the diagram, if any. (5 marks)
- (c) Provide **THREE (3)** examples of entities with conceptual existence. (3 marks)
[Total : 25 marks]
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